

Test Plan:

Project / Feature: SauceDemo E-Commerce Web Application

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1. Scope

The testing scope covers the primary customer shopping journey and critical functionality of the e-commerce web application.

The following areas will be tested:

- User login and logout
- Login validation and error messages
- Product listing and product details
- Product sorting
- Add/remove products from cart
- Shopping cart functionality
- Checkout information
- Order summary and total calculation
- Order completion
- Navigation between application pages
- Basic UI and usability validation
- Cross-browser compatibility
- Regression testing of existing functionality

2. Out of Scope

The following areas are excluded from this test cycle:

- Performance and load testing
- Security penetration testing
- Database-level testing
- Third-party payment gateway validation
- Backend API performance testing
- Production infrastructure testing
- Accessibility compliance certification
- Mobile native application testing

3. Test Objectives

The objectives of testing are to:

- Verify that critical business workflows function as expected.
- Ensure users can successfully log in and complete a purchase.

- Validate positive and negative scenarios.
- Identify functional, UI, and usability defects.
- Verify that incorrect user input produces appropriate validation messages.
- Ensure calculations such as item totals and order totals are accurate.
- Verify that recent changes do not introduce regressions.
- Confirm that the application behaves consistently across supported browsers.
- Provide sufficient test evidence to support the release decision.

4. Test Approach

Testing will follow a risk-based approach, prioritizing critical user workflows.

Functional Testing

- Create positive and negative test cases.
- Validate expected results against actual application behavior.
- Verify field validation and error handling.
- Test boundary and invalid input scenarios.

UI Testing

- Verify layout, buttons, links, labels, and navigation.
- Check that elements are visible and usable.
- Verify dropdowns, sorting controls, and interactive components.

Regression Testing

- Execute a regression suite after significant application changes.
- Prioritize previously failed and business-critical functionality.

Automation Testing

Automated tests will be implemented using:

- Python
- Playwright
- Pytest

Automation will focus primarily on repeatable and high-value regression scenarios such as:

- Login
- Product selection
- Add to cart
- Checkout
- Order completion

Cross-Browser Testing

Critical automated tests will be executed against supported browsers such as:

- Chromium
- Firefox
- WebKit

Defects will be documented with reproduction steps, expected/actual results, screenshots, logs, and test evidence where applicable.

5. Environment

Application: SauceDemo Web Application

Browsers:

- Google Chrome / Chromium
- Mozilla Firefox
- Safari / WebKit

Operating System:

- Windows 10/11
- macOS where available

Automation Framework:

- Playwright
- Pytest
- Python

Test Data:

- Valid user credentials
- Invalid credentials
- Product test data
- Valid checkout information
- Invalid and boundary test data

Test Evidence:

- Pytest HTML reports
- Screenshots
- Playwright traces
- Test execution logs

6. Entry Criteria

Testing can begin when:

- A testable application build is available.
- Required features are deployed to the test environment.

- Test credentials and required test data are available.
- Requirements or acceptance criteria are available.
- Test environment is accessible and stable.
- Critical blocking defects from the previous cycle have been resolved or accepted.

7. Exit Criteria

Testing can be considered complete when:

- All planned critical test cases have been executed.
- All critical and high-priority defects have been resolved or formally accepted.
- No known blocker defects remain.
- Critical user workflows pass successfully.
- Regression testing has been completed.
- Automation tests are passing within the agreed threshold.
- Test results and evidence have been documented.
- A final QA test summary has been prepared.

8. Risks & Mitigation

Risk	Mitigation
Test environment becomes unavailable	Coordinate with the development/DevOps team and maintain a backup test environment where possible.
Requirements are unclear or change frequently	Review requirements with stakeholders and update test cases when requirements change.
Critical defects are discovered late in the testing cycle	Prioritize high-risk and business-critical workflows early in testing.
Test data becomes invalid or unavailable	Maintain a controlled set of reusable test data and credentials.
Automated tests become unstable	Use reliable locators, explicit synchronization, and regular maintenance of the automation suite.
Browser-specific defects are missed	Execute critical scenarios across supported browsers.
Limited testing time	Apply risk-based prioritization and execute smoke/regression tests first.
Defects cannot be reproduced consistently	Capture screenshots, videos, logs, traces, environment details, and exact reproduction steps.